NTNU

Kunnskap for en bedre verden

Department of Computer Science

TDT4252 - Enterprise Architecture for Enterprise Innovation

Semester Report

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Table of Contents

List of Figures **iii**

1 Enterprise Modeling **1**

1.1 Case Description 1

1.1.1 My motivation 1

1.2 Purpose of the models 1

1.2.1 Goals of the model 2

1.3 Design of the model 2

1.4 Enterprise Model 3

1.4.1 4EM 3

1.4.2 Archimate 3

1.5 Perspectives 5

1.5.1 4EM - Goals 5

1.5.2 4EM - Business Process 5

1.5.3 4EM - Actors and Resources 8

1.5.4 Archimate - Motivational View 8

1.5.5 Archimate - Business Process 8

1.5.6 Archimate - Business Actor 10

1.6 Modeling Tools 10

2 Open Service Innovation and Service Design **11**

2.1 Discover 11

2.2 Define 11

2.3 Develop 12

2.4 Experience Points 12

2.5 Blueprint 14

2.6 Reflections 16

3 Business Modeling **16**

3.1 Model . . . 16
3.2 Key Partners . . . 16
3.3 Key Activities . . . 18
3.4 Key Propositions . . . 18
3.5 Customer Relationships . . . 18
3.6 Customer Segments . . . 18
3.7 Channels . . . 18
3.8 Key Resources . . . 19
3.9 Cost Structure . . . 19
3.10 Revenue Streams . . . 19

4 Redesigned Enterprise Model **19**
4.1 New Enterprise Model . . . 19
4.2 Enterprise Modeling and Enterprise Architecture . . . 21

5 Assessment **22**
5.1 Evaluation of Enterprise Model . . . 22
5.1.1 Physical Quality . . . 22
5.1.2 Empirical Quality . . . 22
5.1.3 Syntactic Quality . . . 23
5.1.4 Semantic Quality . . . 23
5.1.5 Pragmatic Quality . . . 24
5.1.6 Social Quality . . . 24
5.1.7 Deontic Quality . . . 24
5.1.8 Overall . . . 24
5.2 Reflections . . . 25

Bibliography **26**

List of Figures

1 Enterprise model in 4EM 4

2 Enterprise model in Archimate 6

3 Goals model in 4EM 7

4 Business process model in 4EM 7

5 Actors and resources model in 4EM 8

6 Goals model in Archimate 9

7 Business process model in Archimate 9

8 Business actor model in Archimate 10

9 Company contacting student organisation before 13

10 Company contacting student organisation after 13

11 Student organisation getting contacted by company before . . . 13

12 Student organisation getting contacted by company after 14

13 Service Blueprint with new website 15

14 Business model 17

15 New enterprise model in 4EM 20

16 Example of different types of lines 23

17 Crossover of relations 23

1 Enterprise Modeling

This section will start off by presenting the chosen case, my motivation for choosing this case, and possible problems detected with today's solution. Additionally, I will show some chosen aspects of the case in both 4EM and Archimate.

1.1 Case Description

Due to the rapid rise of technological needs in the past years, IT students are some of the most sought after in the industry. This demand can be seen through events such as itDAGENE[3] and Tech Talks[5], where companies pay a lot of money to come up to Trondheim to speak to IT students. Furthermore, IT student organizations, like Online[6], get paid to allow companies to hold presentations for their members. This allows students to get to know companies from all over the world and learn what their values are. All of this is great for the most sought after students, but how can a company guarantee themselves that the best students apply for a job with them and how should they go forward to be able to make this guarantee?

For my case I have chosen an arbitrary IT company, who is looking to hire the best IT students from NTNU to their company. In this report I will present some aspects of how this IT company goes forward in marketing themselves to students, and then present my innovation plan for how companies can better go forwards in guaranteeing themselves the best students.

1.1.1 My motivation

As an IT student, my main motivation for choosing this case is that I wanted to learn how IT companies go forward in finding students for their graduate and summer intern positions, and try to figure out what type of marketing strategies they use to get students interested in them.

1.2 Purpose of the models

The most important part of an IT company is the employees of the company, since they are the ones creating value. The better the employees are, the more money a company can make. This is why IT companies try to get the best students, to secure the highest possible gain.

There are many manual tasks that are involved when trying to reach out to students, so by modeling some of the processes that happen when reaching out we might find possibilities for innovations that save both time and money. This would lead to a higher marketing budget and more time to find the best students.

It is also important to map out the users and main beneficiaries of the models before starting the modelling process, to make sure that the model includes everything that is relevant and important to them. In this case the main beneficiaries would be student organisations and IT companies looking to contact students, while the users of the model would be recruiters in the aforementioned IT companies and volunteers in the student organisations that are responsible for company communication.

1.2.1 Goals of the model

According to the Sandkuhl's book [7] we create enterprise models to get an AS-IS overview of the current situation in an enterprise, and by utilising the model one can get a better understanding of where changes and improvements can be made. Therefore the main goals of this enterprise model can be defined as:

- Get a good understanding of how IT companies contact students
  - Sub-goal: Find places to innovate and make changes, to make this process better
- Visualise the processes involved in student recruiting
  - Sub-goal: Give the users an easy to understand overview of the current situation

With both of these goals one can also define the enterprise model's success criteria, as the enterprise model will end up successful if the goals above are fulfilled.

- Goal 1:
  - It should be clear where changes need to be made and where innovation can be done.
- Goal 2:
  - All the targeted users should manage to understand all aspects of how communication between IT companies and student organisations are done today.

1.3 Design of the model

Since the main purpose of the model is to find manual tasks in the process of companies communicating to student organisations that can be innovated on, the models design puts focus on the business processes involved. For

the reason that the business process models give a good insight into what actually happens between the two parties and makes it somewhat easier to see optimisation opportunities. The model also includes a goals perspective and organisational structure, as to give an indication on why one should innovate in this field and who these changes would affect the most.

1.4 Enterprise Model

Both the enterprise models, in 4EM and Archimate, include perspectives that gives the same insights into the enterprise. The purpose of this was to find out which language was best suited for this project, and that these perspectives were considered to give the best overview over potential inefficiencies in the enterprise.

1.4.1 4EM

The 4EM enterprise model (seen in figure 1), includes the Goals (figure 3), Business Process (figure 4), and Actors and Resources perspectives (figure 5). Between the different perspectives one can also find some relationships.

- Goal model:
  - The goal model is connected to the business process model through Goal 3, "Have more company presentations". Goal 3 motivates a company to contact more student organisations, as this will further increase the chances of them fulfilling their ultimate goal, which is to recruit more students.
- Actors and Resources model:
  - The Actors and Resources model is connected to both the Goal model and Business Process model. The marketer role within the marketing department is connected to Goal 4, "Increase marketing budget", as they would be partly responsible for increasing the marketing budget of an enterprise. The recruiter, in the HR Department, is connected to the Business Process model as they have the responsibility to set up and contact student organisations. Furthermore, the arbitrary company XYZ defines their main goal which is to recruit more NTNU students.

1.4.2 Archimate

The Archimate enterprise model (seen in figure 2) was heavily inspired by the 4EM model, as I tried to translate the different 4EM perspectives into the

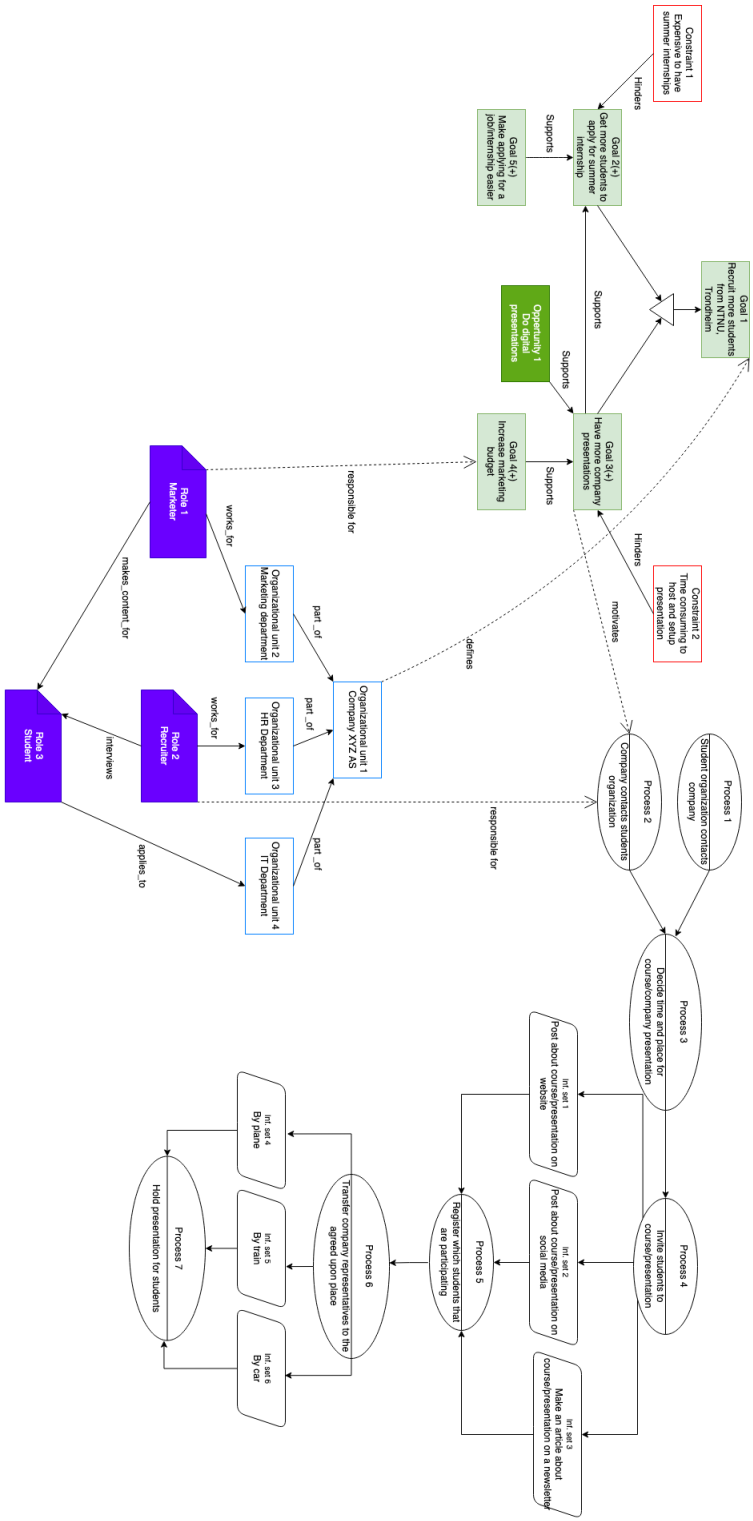


Figure 1: Enterprise model in 4EM

perspectives that existed in Archimate. This resulted in the enterprise model being built up of a Motivational (figure 6), a Business Process (figure 7), and a Business Actor perspective (figure 8). As with the 4EM enterprise model, this model also included some internal relationships between the different perspectives.

- Motivational model:
  - The Motivational model is connected to both the Business Actor and Business Process models, as the goal model directly motivates the enterprise to go out and communicate with student organisations to get more NTNU students.
- Business Actor model:
  - The Recruiter within the Business Actor model is connected to the Business Process model, as they are responsible for contacting the student organisations.

1.5 Perspectives

This section will further elaborate on the different perspectives of the Enterprise Models from both 4EM and Archimate.

1.5.1 4EM - Goals

The purpose of the goal model (figure 3) is to describe what the enterprise is pursuing, what they are striving towards [7]. Goals can also have goals that support them, which gives a good indication in what an enterprise should do to fulfill their main goal. There are also some constraints that hinder the realisation of a goal.

1.5.2 4EM - Business Process

The purpose of the Business Process model (figure 4) is to give an insightful overview of how a process inside the enterprise is handled [7]. In this case, I have chosen to model how a usual company presentation is done. I chose this because, from my experience, this is the best way a company can reach out to students and is usually the most effective for getting students to apply for a job at your company.

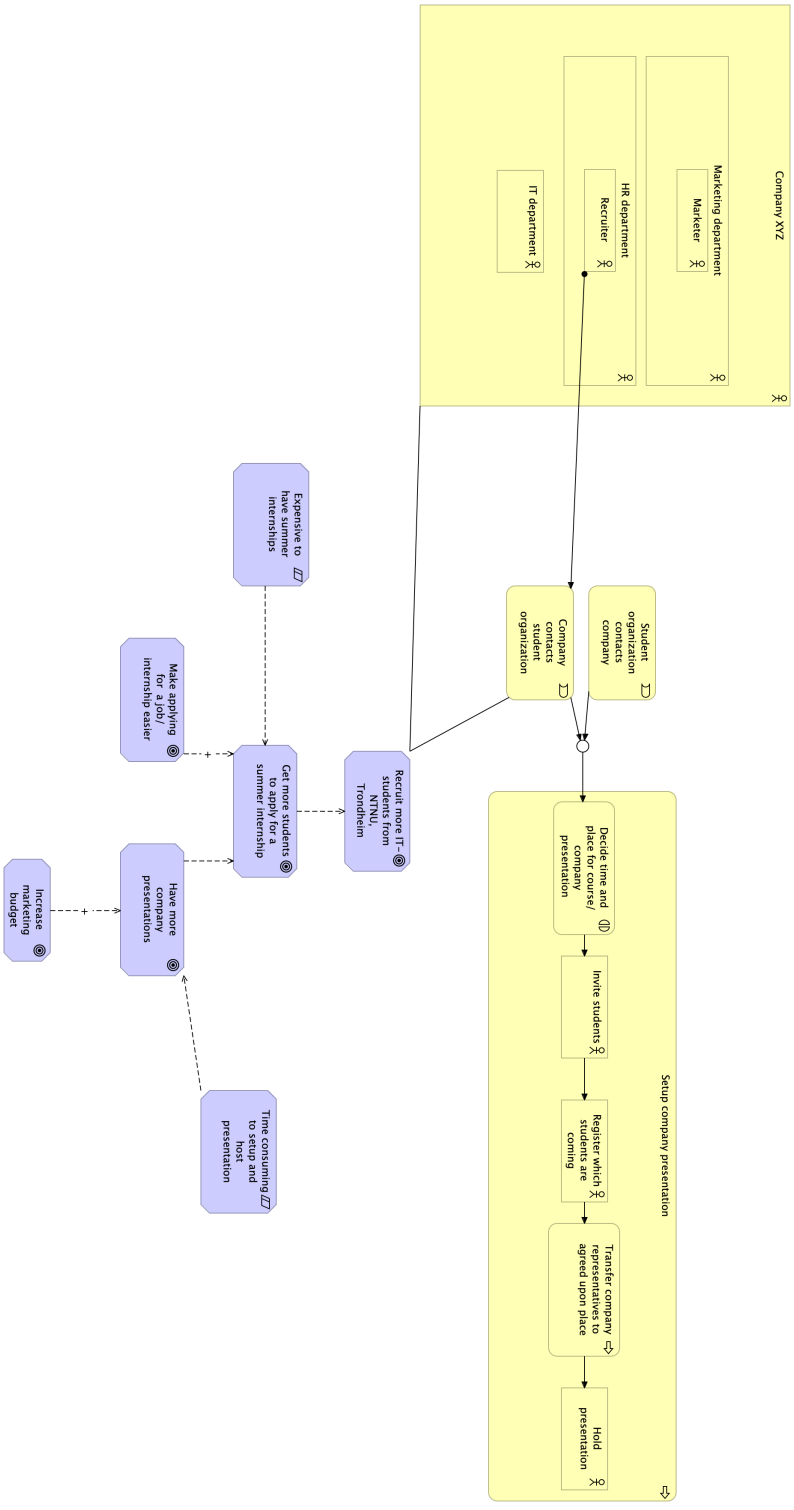


Figure 2: Enterprise model in Archimate

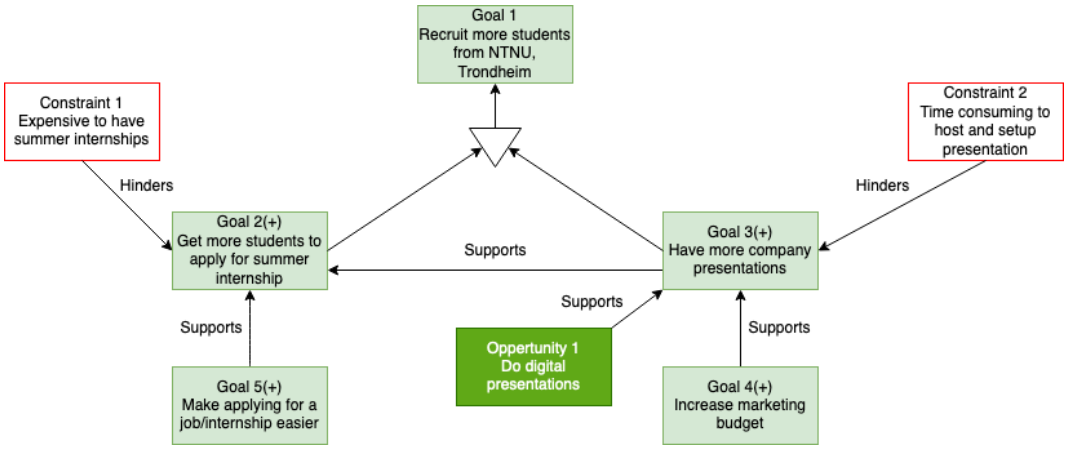


Figure 3: Goals model in 4EM

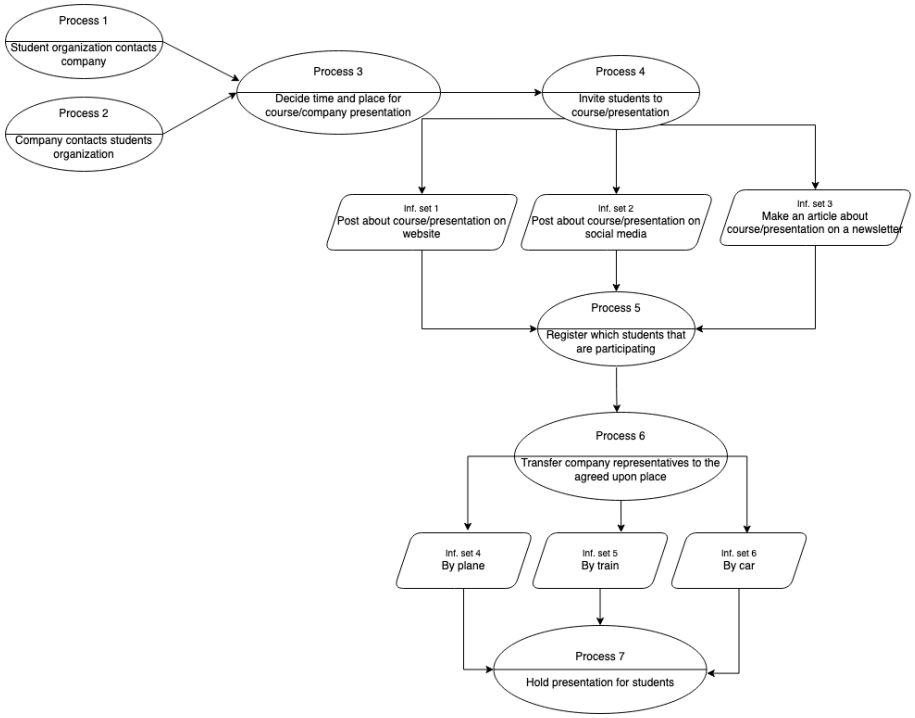


Figure 4: Business process model in 4EM

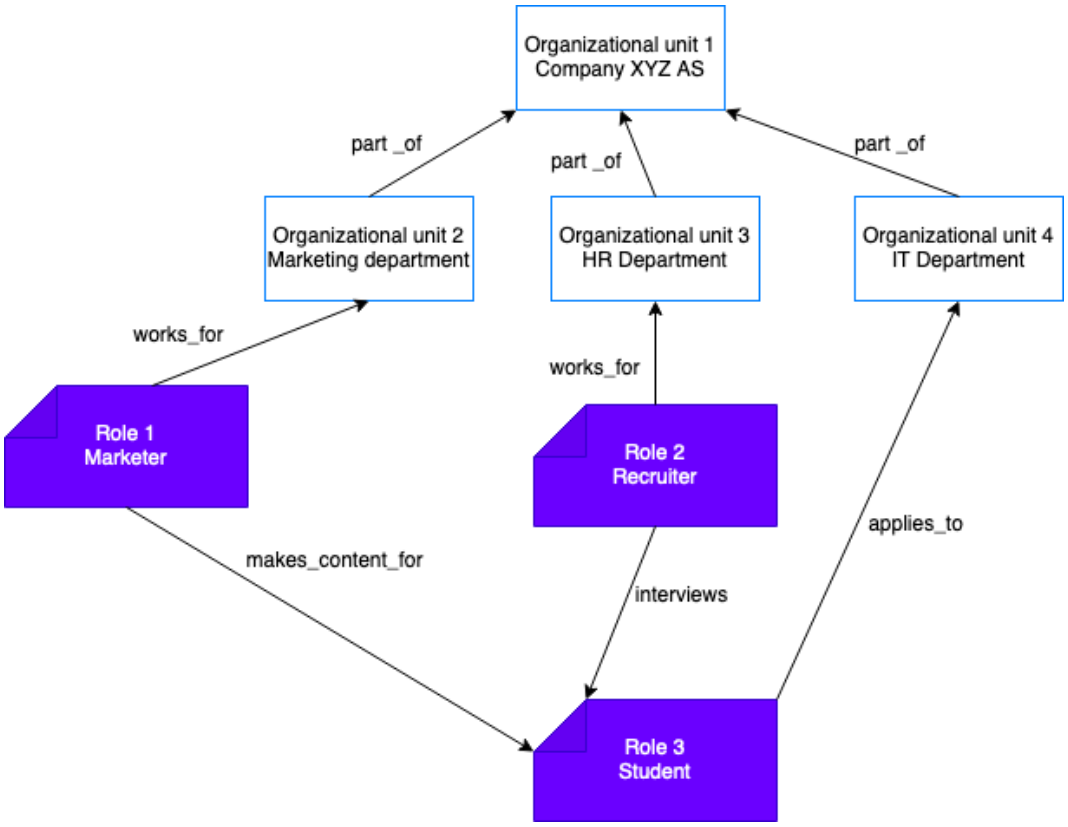


Figure 5: Actors and resources model in 4EM

1.5.3 4EM - Actors and Resources

We create Actors and Resource models (figure 5) to show how an organisation is structured, and how different entities inside the organisation are related to each other [7]. In this case, I have only modeled the parts of an enterprise that have some connection to the process of recruiting students.

1.5.4 Archimate - Motivational View

This motivational view (figure 6) has the same purpose as the 4EM Goals model, but is structured in a slightly different fashion.

1.5.5 Archimate - Business Process

The Business Process model (figure 7) has also modeled the same process as in 4EM, company presentation, but in Archimate I decided to put all the administrative setup elements inside it's own element. This makes it easy to see what triggers the process and what happens inside the process.

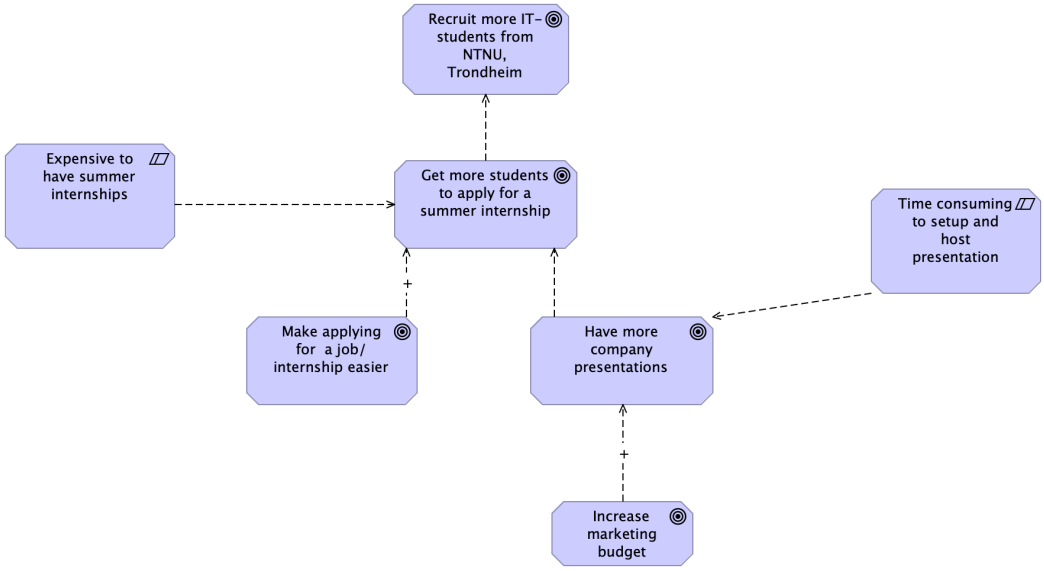


Figure 6: Goals model in Archimate

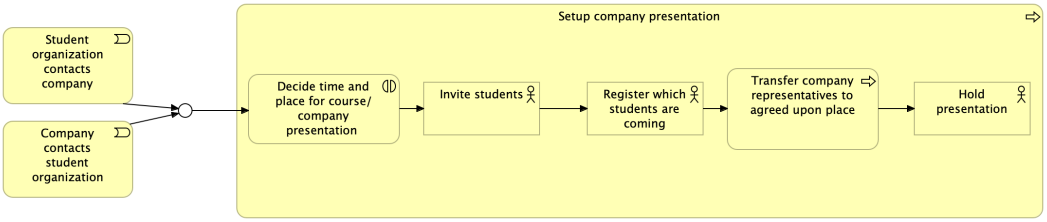


Figure 7: Business process model in Archimate

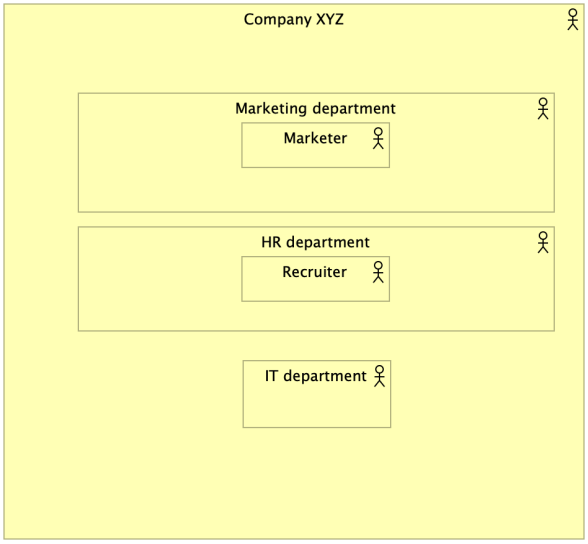


Figure 8: Business actor model in Archimate

1.5.6 Archimate - Business Actor

The Business Actor model (figure 8) has again the same purpose as in 4EM, but gives a very simple overview of how the enterprise is structured.

1.6 Modeling Tools

Before modeling, I read up on how to use both 4EM and Archimate, to try and find which modeling language would work best in this project. After a while I figured that it would be very hard to work out which one would be best, without having any experience in any of them. So I started out with 4EM, as it seemed similar to other modeling languages I have used before, like UML. After creating the enterprise model in both languages, I came to the conclusion that I wanted to continue using 4EM. Archimate had a lot of features and an intuitive program (Archi[1]) to create the models, but all the different elements that one could use in a model felt very unnatural to me. I am used to simple modeling tools, where you only deal with small set of different elements when creating a perspective. This is what I liked about 4EM, since in each perspective you had a very good understanding of what elements you should use and how these connect to each other. Also, I feel like 4EM is easier to understand since you can read what each element and connection mean, while in Archimate you have to understand what all the different element tags mean to get a full understanding of the model.

[1] https://www.archimatetool.com/

2 Open Service Innovation and Service Design

In this section, I will present the chosen service innovation method for this project and how I used it to innovate on the enterprise. I chose the Double Diamond Design Process, as described by *The Design Council* [2], as this seemed like a largely used and respected process.

The Double Diamond Design Process includes four steps: Discover, Define, Develop, and Deliver. I decided to leave the Deliver step out of this project, as this step involves testing out the solutions which is not in the scope of this course.

2.1 Discover

The Discover phase is all about discovering problems with the current enterprise and finding places to innovate. By analysing the models I previously created and through conversations with students, in my student organisation, that have the responsibility for keeping in contact with outside companies, I discovered the following areas that should be further looked into.

- Finding new student organisations that have the students you are looking for. Usually, a company finds student organisations by having employees that have been a part of the organisation before or the company has their own department for this purpose. This restricts the companies to a small set of student organisations, or they have to invest a lot of money and time into finding other organisations.
- Setting up and holding a presentation takes a lot of time and resources. Currently, this is a very manual job with a lot of emails back-and-forth. Also after the presentations all the companies have their own specific method to get billed, so the students have to use a lot of their spare time to find out how they have to structure their invoices.

Now that we have discovered and explored some of the inefficiencies in the current model, we can continue to the Define step where we will try to find some opportunities that we can utilise.

2.2 Define

The Define step involves using the data we got from the Discover step, and using it to find opportunities. The Discover phase ended up with two discovered problems, it's hard for a company to reach out to all student organisations that have students they want and it takes a lot of manual work to set up company presentations. With this I started trying to find something that could make these processes easier, and I came up with the following idea:

- A website where student organisations and companies can find each other based on common interests. By using one platform, both the companies and student organisations can more easily find each other. Furthermore, some standards can be implemented so that payment and other administrative processes goes a lot easier.

With an idea defined, we can go to the Develop step where I will further outline how a product like this can be implemented.

2.3 Develop

To create this application you would need some software engineers, as we are innovating for an IT company they can take out some of their software engineers from their staff and put them on this project. The project should follow a development framework, like SCRUM[2], that force the developers to regularly test out the application on the target audience. This is because they need to ensure themselves that the users they are going to sell this product to, actually want to use it. The development will also follow the principles of open innovation, as this would allow for pre-made software that is well documented and tested to be used, which will lead to faster time to market.

The application would be a platform where all companies and student organisations that want to use it, can do. The reasoning for not only restricting the application to my company, is because this is an application many companies would have use for and therefore one could earn a lot of money on a subscription based service model, further described in the business model.

2.4 Experience Points

Here I will present the process a company and student organisation goes through, before and after the innovation. How the process looks from the company's view can be seen in figures 9 and 10. At first glance it might seem that the company suddenly has more things to do, but in reality each box in the first figure includes a lot of manual work that is lost when using the website. For example, the "Find student organisation" point can take a lot of money and resources, as discussed before, but with the website a simple search is enough. Furthermore, many of the other points will take a lot of emails back-and-forth to discuss times and prices, but the website will manage to do this for them. In the student organisations case, seen in figures 11 and 12, one can see a significant difference. After the registering on the website, a student organisation only needs to accept applications from companies and find a room for them to have their presentation in.

[2] https://www.scrum.org/

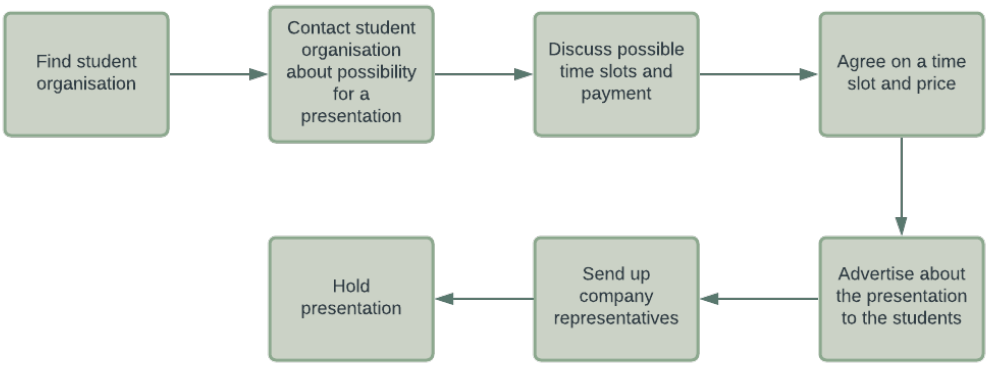


Figure 9: Company contacting student organisation before

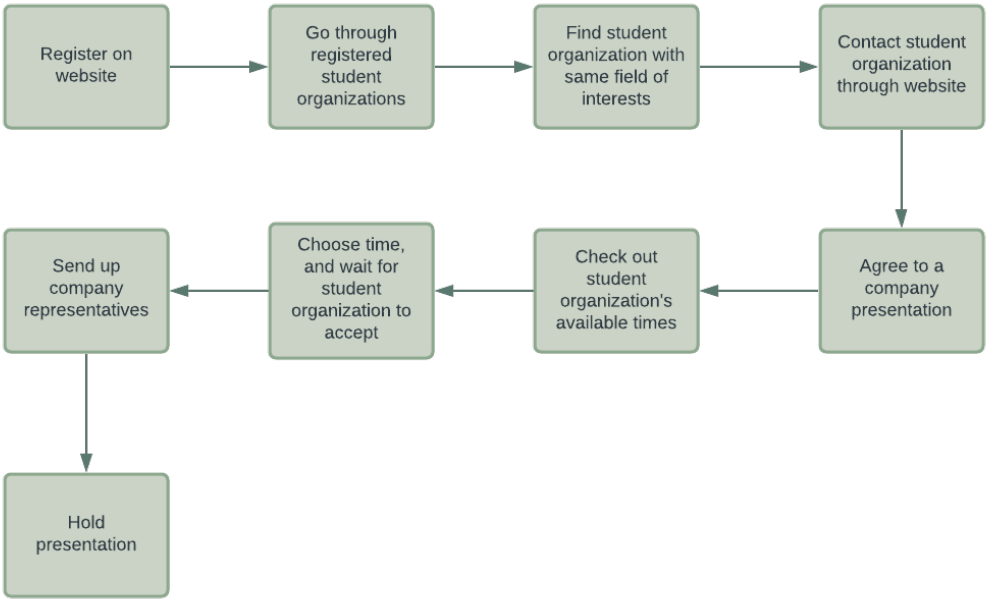


Figure 10: Company contacting student organisation after

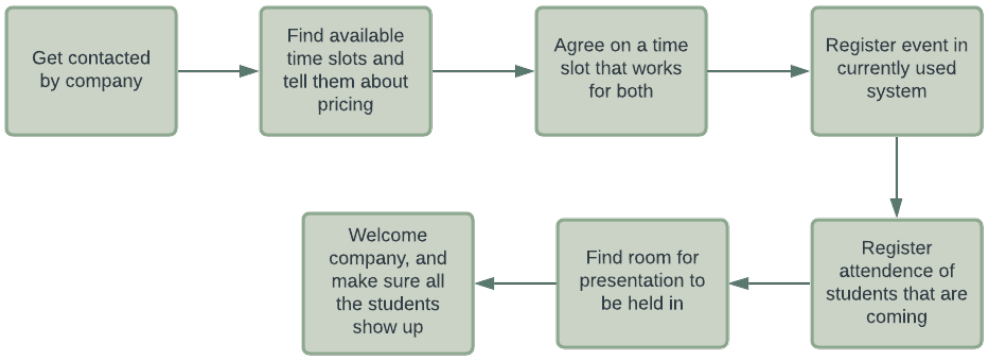


Figure 11: Student organisation getting contacted by company before

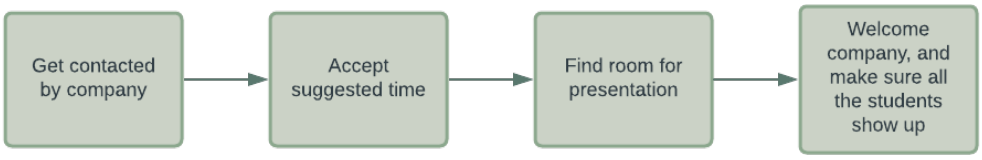


Figure 12: Student organisation getting contacted by company after

2.5 Blueprint

The Service Blueprint can be seen in figure 13. Here we see that a lot of the work that the employees of the company had to do before, has been moved to a backend service that does the work for them. The only tasks a company employee has to do themselves, before the presentation, is to log into the website and find a student organisation that have students within their field. It is important that this website is easy to use and does it job in as little time as possible, since if not people will go back to using the manual methods. With the new innovation a company would not need a whole team to find new students that they want to recruit, and only need to put off a little time to browse through the website and find potential matches.

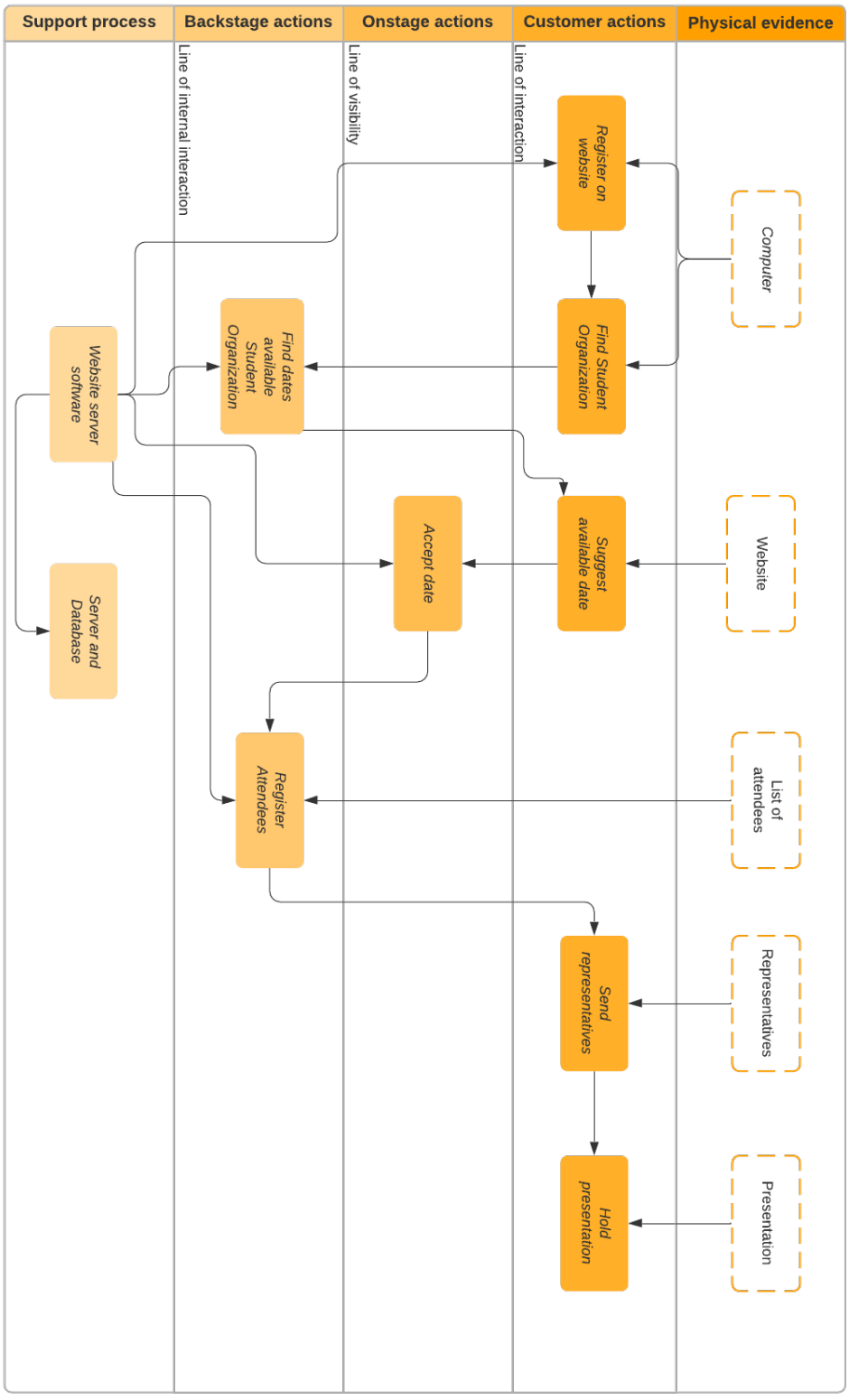


Figure 13: Service Blueprint with new website

2.6 Reflections

Innovation is the most important part of growing a company, as it can provide new market opportunities and attract customers from other other companies to you. A lot of companies keep their customers by brand loyalty or them having a good reputation, but when looking at new companies, like Tesla[3] and Amazon[4], that have created new platforms through innovation you see tremendous growth in a very short time span.

By using innovation modelling tools, such as experience points and blueprints, one can easily see the impact of a new service or product, by comparing before and after. Compared to enterprise models, these types of models give much more detailed insight into how the innovations affect the main beneficiaries of the product, while enterprise models gives a better overview of how the product will have an impact on the organisational structure of an enterprise. When using both of these modeling techniques together, one can see both the inside and outside effects of a new innovation.

3 Business Modeling

This section will present the business model, as seen in 14, and explain each part of the model in further detail.

3.1 Model

The Business Model Canvas includes nine different fields, which together give a very good overview of how a business will manage to earn it's money. The nine fields include: Key partners, Key activities, Key propositions, Customer relationships, Customer segments, Channels, Key resources, Cost Structure, and Revenue Streams. I chose this model as I found it easy to use, and it gave me a lot of playing room for how I should structure my business after having implemented the new innovation.

3.2 Key Partners

The purpose of the Key Partners field is to map out the people and other parties that make this proposed business model work. My key partners for this business model include all the ones responsible for setting up and creating the innovation product, as well as the intended main beneficiaries for the new application.

[3] https://www.tesla.com/
[4] https://www.amazon.com/

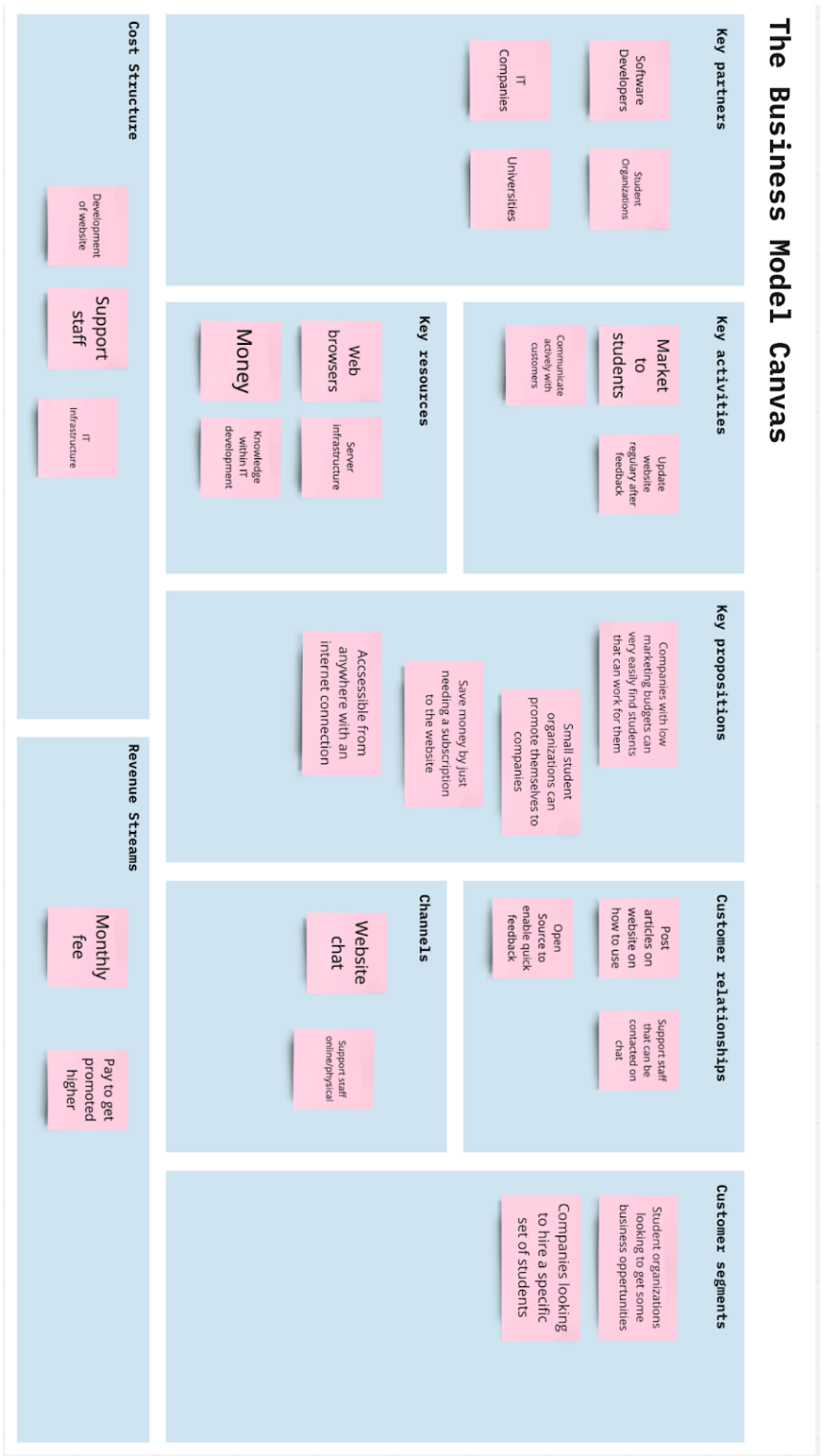


Figure 14: Business model

3.3 Key Activities

The Key activities include what the company must do to make the business model work. Here I have included that they should market first to students and student organisations. The reasoning behind this is that if all the student organisations are found on the website, companies will after a while follow suit. It is also important to listen to the feedback from the customers, and update the website after their needs.

3.4 Key Propositions

The Key Propositions (or Value Propositions) are what value I will create for my customers. I think the model describes them pretty well, but to sum them up the main value propositions that this application will bring are that it will save a lot of time and money for companies that manually looked for student organisations before and it will give smaller student organisations the ability to be seen and organise company presentations in an easy way.

3.5 Customer Relationships

The Customer Relationships include how the company should keep in contact with it's customers. In my model, I have imagined that the company implements very standard relationship strategies as they have been proven to work. By posting articles, people can work out how to use the websites themselves, but if they still have problems they can contact the company or post an issue to the Open Source project.

3.6 Customer Segments

The Customer Segments are a description of the intended customers that the company wants to reach out to. In this case, it would be student organisations and other companies.

3.7 Channels

The company needs to reach out to potential customer segments and present the key propositions they have created, this is done through a set of Channels. Here I have included the website chat, as customer segments that are not customers yet could ask questions, and with a support staff that would also work as a sales force where they go out to potential customers and present the application.

3.8 Key Resources

The Key Resources are the most important resources needed by the company to make this business model work. Here I have included that there is needed a lot of competency within IT and IT-architecture, this is to manage to create the application, as well as investment money to fund the project.

3.9 Cost Structure

The Cost Structure are the costs which are a part of a creating the business model. The main costs in this model would be the ones related to developing the application, as well as to the support staff who also sell the application to the customer segments.

3.10 Revenue Streams

The Revenue Streams would be methods that this business model can earn revenue for the company. The main revenue stream I imagined would be possible for this business model would be having a subscription model, where the customers pay a monthly fee. The monthly fee would differ based on the economic status of the customers, so a student organisation would pay less than a large company. One could also allow companies, or student organisations, to pay to get listed higher in the result lists.

4 Redesigned Enterprise Model

In this section I will present how the new innovation has changed the original enterprise model (as seen in figure 1), and reflect on the relationship between Enterprise Modeling and Enterprise Architecture.

4.1 New Enterprise Model

The new Enterprise model can be found in figure 15. I found it quite hard to implement the new innovation into the enterprise model since this innovation helps more "behind the scenes", but there are some changes that are notable.

- Goal model:
  - I added a new goal (#6), which is to meet more student organisations. With the innovation, they can more easily find other smaller student organisations at NTNU which have a study program that fits their business.

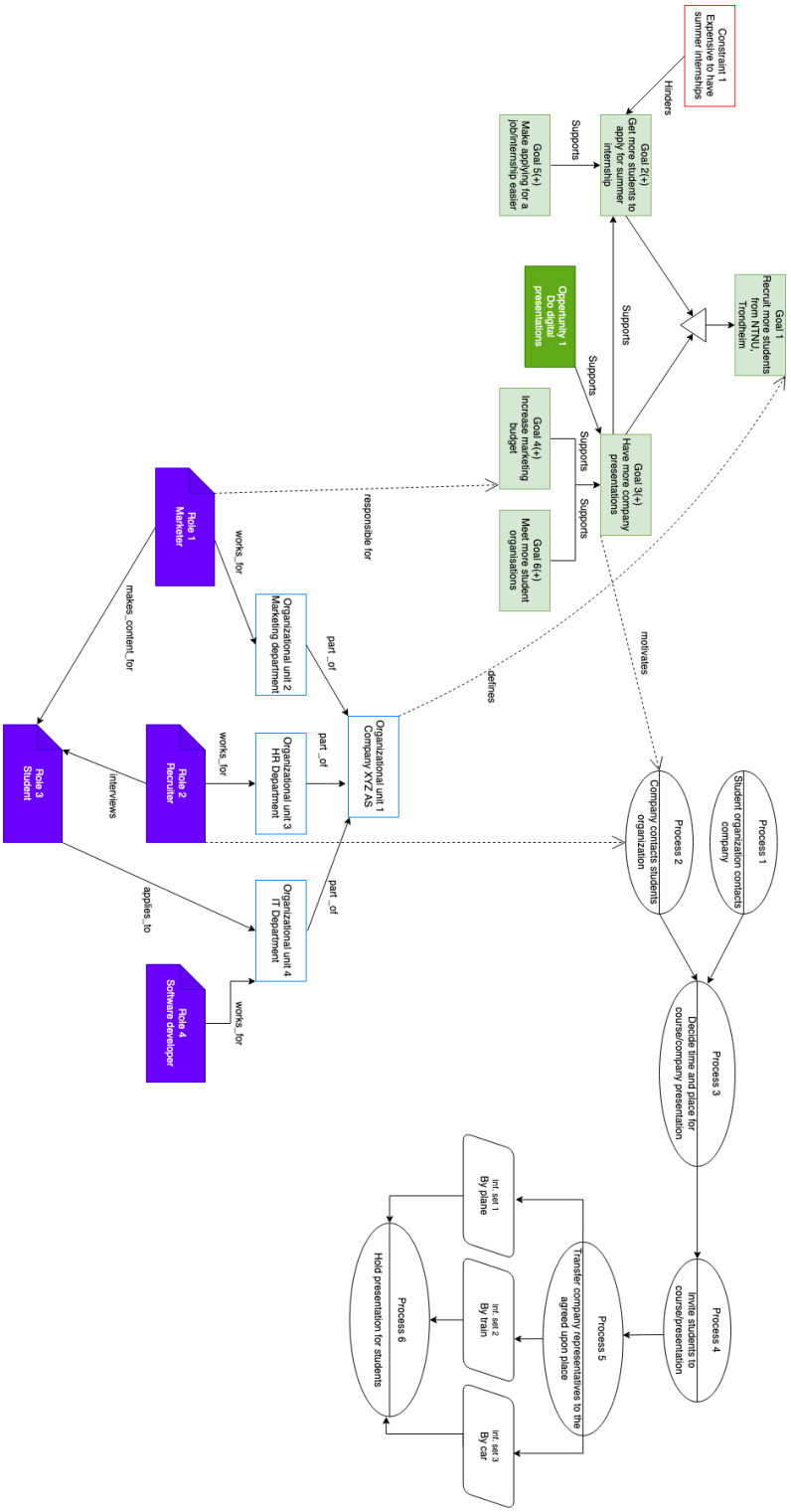


Figure 15: New enterprise model in 4EM

  - I also removed *Constraint 2*, which was that it was time consuming to host and setup a company presentation. Now, with the application, it takes very little time and resources to find student organisations and I no longer consider it a constraint.

- Actors and Resources model:
  - I added the software developer role under the IT department, as they now have a critical role in developing the new application.

- Business Process model:
  - The process of setting up a company presentation has been shortened drastically, as most of the administrative work, such as planning when the presentation should take place and payment, is done by the new application.

The goals of the new Enterprise Model have not changed from before (1.2.1), as new innovations can still be made in this enterprise. With this new model, the original main beneficiaries should be able to see the improvement made on the processes within the enterprise and the effects the changes have had on the organisation.

4.2 Enterprise Modeling and Enterprise Architecture

Here I will discuss the relationship between Enterprise Modeling and Enterprise Architecture, but before that we should define what the different terms mean.

Enterprise Modeling Enterprise Modeling, as defined by Sandkuhl's *et al.* book [7], is "a technique that helps to capture the different elements and structures of an enterprise as well as to visualize the inter-dependencies between the elements". In other words, it's a way to visualize an enterprise.

Enterprise Architecture Enterprise Architecture is "coherent whole of principles, methods, and models that are used in the design and realisation of an enterprise's organisational structure, business processes, information systems, and infrastructure"[1]. Enterprise Architecture tries to capture the essentials of the business, and sets a focus on the IT systems.

From their definitions they both seem very similar, but the main difference is that Enterprise Architecture focuses more on the technological aspects of an enterprise while Enterprise Modeling focuses on how an enterprise is built up and how the organisation communicates internally. Many elements and relations in my enterprise model (figure 15) are there to aid in technological innovation, but since the model mainly only presents how the enterprise is organised I deem it an Enterprise Model.

5 Assessment

In this section I will evaluate the new enterprise model based on the SEQUAL framework, from Krogstie's book [4]. Furthermore, I will reflect on the process I have been through this semester.

5.1 Evaluation of Enterprise Model

Here we will try to evaluate the new Enterprise Model (figure 15), by using the SEQUAL framework. I chose the SEQUAL framework as it covers many different aspects of the model, and I consider it the one of the best ways to evaluate an Enterprise Model. First, we need to remember the original main goals of the model, as we want to find out if the model we ended up with fulfilled these goals. The goals were:

- Get a good understanding of how IT companies contact students
- Visualise the processes involved in student recruiting

Now we can go through the different parts of the SEQUAL framework, and evaluate the model.

5.1.1 Physical Quality

The Physical Quality of the model is based on how available it is to the relevant Actors (main beneficiaries) and if it is up-to-date with the latest enterprise changes.

The 4EM models are created in the online modeling tool *draw.io*[5], and stored on Google Drive[6]. As long as the Actors have a Google account, and I send them an invitation link to the models they can easily access them. A perk of using *draw.io* is that changes are made in real time, so if I change the model then an Actor can see the change in a matter of seconds. The models are also as up-to-date as they can be, since they were based on the new innovations presented in section 2. With this we can say that the models pass the Physical Quality criteria.

5.1.2 Empirical Quality

The Empirical Quality of a model is measured on the models' aesthetics, *i.e.* the readability and consistency of the model. Because of how 4EM is set up I would consider it quite easy to read the model and understand what the

[5] https://app.diagrams.net/
[6] https://www.google.com/drive/

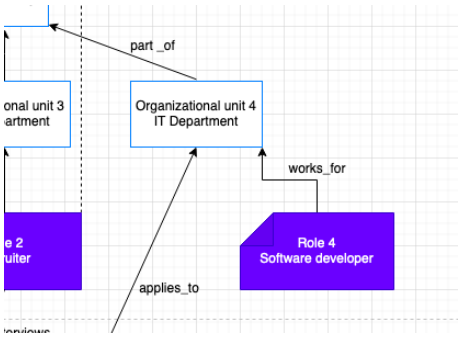


Figure 16: Example of different types of lines

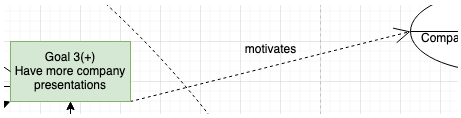


Figure 17: Crossover of relations

different elements do, without having to know 4EM from before. This was one of the reasons I chose 4EM. The model also keeps good consistency as each perspective uses their own elements, with no mixing of what the different elements do. However, I have connected some elements with skewed lines while others are connected with straight lines (seen in figure 16). This does not directly impact the readability of the model, but can cause some confusions when reading the model. There are also some relations in the model that cross over each other, example seen in figure 17, which can affect the model's readability. But this is very rare, and I would not say that it affects the readability of the model because it does not block any information.

I would say that the models pass the Empirical Quality criteria and also helps in fulfilling both goals, as the good readability aids in visualising the student recruiting process and in giving an understanding of the communication between companies and student organisations.

5.1.3 Syntactic Quality

The Syntactic Quality of a model is how correct the model is created, based on the rules of the modeling language. The model follows the rules defined in the *Tackling Business Challenges with the 4EM Method* book, and I tried to not take any liberties when creating my model. From my understanding of the 4EM rules, the model passes this criteria.

5.1.4 Semantic Quality

The Semantic Quality measures if the created model actually fits within the original domain of the project, and that there are no elements in the model that shouldn't be there. The domain is how a company contacts a student

organisation, and all the presented elements and perspectives in the model are important parts of this process. With this, the model would pass the Semantic Quality criteria.

5.1.5 Pragmatic Quality

The Pragmatic Quality tells us if the Actors understands what the model expresses. As discussed before, 4EM is easy to understand even for people who do not know much about modeling from before, and with a textual description of what the purposes of the model is, the Actors should be able to understand what the model is trying to express.

5.1.6 Social Quality

The Social Quality of the model is defined by if the Actors agree with the quality of the model. This is hard to say from the companies' perspective as I have no direct contacts with employees that have this responsibility, only an understanding of how this process is done. I have discussed the case and model with a classmate who has the responsibility of contacting companies in my student organisation and they agreed with the quality of the model, but said I might have oversimplified the company presentation process as there is a lot that goes into this process. This leads to the Social Quality criteria being somewhat passed, but there are still a lot of aspects of the Social Quality I was not able to test.

5.1.7 Deontic Quality

The Deontic Quality of the model measures if the model actually fulfilled it's original goals. As came forward in the Empirical Quality section, I would consider the original goals of the model to be fulfilled. Since the models gave me a good understanding of the communication process between students and student organisations, as well as giving me a good visual over a part of the student recruiting process. The models also gave a good indication on where I should focus on innovating. I would therefore say that the model passes the Deontic Quality criteria.

5.1.8 Overall

Aside from some problems in the Empirical Quality and that I was not able to test all aspects of the Social Quality, I would say that the model is good according to the SEQUAL framework. There are always some room for improvements, and I will take my mistakes into account the next time I am going through this process.

5.2 Reflections

I have learned a lot throughout this semester, but not without any hurdles. This is my first time modeling enterprises and I struggled a lot to start out with. I chose 4EM as it had the most similar syntax to other modeling languages that I have used before, so using that went good, but finding out which part of the enterprise is important to model and which parts I could leave out took a long time to figure out. It was also hard to choose which perspectives I should choose, but I got some help from one of the student assistants and ended up being very happy with the perspectives I chose.

The innovation part of the process was very interesting. I have done similar processes in other courses before, but not any ones that are as well documented as we have done this semester. It was a real example of "trust the process". It was also really interesting to create a business model, as there are so many factors one has to think about.

I found it hard to redesign the enterprise model, and I did not give me enough time to do it. This resulted in an adequate enterprise model that does not really take all the effects of the new innovation into play. For next time I have to give myself a lot more time to think and redesign, to ensure myself that the design actually uses the innovation.

All in all, this has been a very fun project where I have learned a lot, both through failing and reading theory. This project has given me a lot of insight into how innovation actually works, and I will definitely use theories learnt here if I ever decide to try to innovate on a solution.

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